Resting-State fMRI Data Quality and Analysis in the WAND 7T Dataset

A Comparative Evaluation of Preprocessing Pipelines

Darsh Kodwani
Royal Holloway, University of London
darsh.kodwani@rhul.ac.uk

Narender Ramnani
Royal Holloway, University of London
n.ramnani@rhul.ac.uk

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Abstract

Placeholder — write abstract here.

1 Introduction & Context

1.1 7T fMRI: Opportunities and Challenges

1.2 The WAND Dataset

The Welsh Advanced Neuroimaging Database (WAND) is a publicaly available multimodal nueroimaging dataset. It was releades by the Cardiff University Brain Research Imaging Centre (CUBRIC) in 2025. The dataset was acquired on 178 healthy adult participants, aging between 18 and 63, who volunteered for the study. In total, 8 imaging sessions were acquired (but not every participated in all sessions). The sessions were across various modalities and are explained below:

- **Session 1:** Structural MRI (T1-weighted, T2-weighted, MP2RAGE)
- **Session 2:** Resting-state fMRI (7T, 1.5 mm isotropic, TR = 1.5 s, multiband acceleration)
- **Session 3:** Task-based fMRI (7T, same parameters as resting-state)
- **Session 4:** Diffusion MRI (7T, multi-shell acquisition)
- **Session 5:** Arterial Spin Labelling (ASL) perfusion imaging
- **Session 6:** Magnetic Resonance Spectroscopy (MRS)
- **Session 7:** Physiological recordings (cardiac, respiratory)
- **Session 8:** Behavioral and cognitive assessments

1.3 Aims

2 Pre-processing and Data Quality

2.1 Image Quality Metrics

The temporal signal-to-noise ratio (tSNR) is defined voxel-wise as:

$$\mathrm{tSNR}(\mathbf{x}) = \frac{\bar{S}(\mathbf{x})}{\sigma_t(\mathbf{x})} \tag{1}$$

where $\bar{S}(\mathbf{x})$ is the temporal mean and $\sigma_t(\mathbf{x})$ is the temporal standard deviation of the BOLD signal at voxel $\mathbf{x}$. The group-level metric reported is the median tSNR across all brain voxels.

2.2 Motion Correction

2.3 Brain Masking

2.4 Group-Level Quality Assessment

2.5 Preprocessing Pipeline Comparison

Table 1: tSNR for subject sub-01187 under four preprocessing conditions. MRIQC value serves as the reference standard.

Condition	**tSNR**	**Δ vs MRIQC**
Raw + simple intensity mask (baseline)	21.18	-7.43 (-26.0%)
Raw + nilearn EPI mask	25.78	-2.83 (-9.9%)
Moco + simple intensity mask	26.85	-1.77 (-6.2%)
Moco + nilearn EPI mask	31.45	$+2.83$ ($+9.9\%$)
MRIQC reference	28.62	—

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